Temperature and particle motion

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a drawing or explanation of particles to distinguish average motion from total energy.

SUCCESS CRITERIA - I can...

1. Describe temperature in terms of microscopic motion.
2. Use kelvin in proportional temperature comparisons.
3. Avoid treating a single particle speed as the temperature.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “kelvin” and “particle” to explain your answer. Sentence starters are optional.

kelvin: the absolute temperature scale used in thermal ratios

particle: a small constituent, such as an atom or molecule

temperature: a measure of thermal state; it is not an energy amount

01 Warm-up: bring back an earlier idea

Sunlight carries energy across **3** million km of nearly empty space in a simplified example. **Name the transfer mechanism** and **explain why matter** is unnecessary.

Sentence starter: I think ... . I can point to ..., which shows ..., so ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

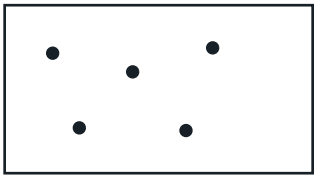
Reference: $K_{avg} \propto T$; $T(K) = T(°C) + \mathbf{273.15}$.
Ideal gas: mean translational kinetic energy per particle scales with absolute temperature. ∝ means proportional to.

Worked reasoning: In a simplified description of a gas, a higher temperature means the particles have greater average kinetic energy from random motion. Individual particles still have a range of speeds.

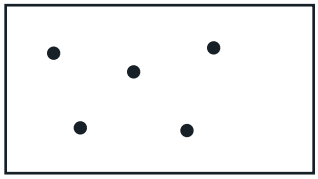
Your turn: Draw two equal-size samples using dots and arrows; **label arrows as particle velocities** and make the warmer sample's average arrow length greater.

Sentence starter: Both ... . They differ in ... because ... .

Sample A



Sample B



Add velocity arrows; use lengths to show a range of speeds.

Marzano check - circle: ① ② ③

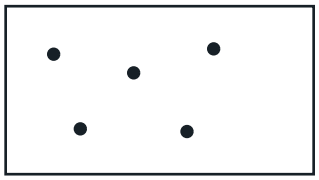
03 Read and explain the diagram or equation

Each box represents a sample of the same ideal gas. One is cooler and one warmer. **Add particle-velocity arrows** with a range of lengths in each box. Make the warmer sample’s average kinetic energy greater, without making every particle identical.

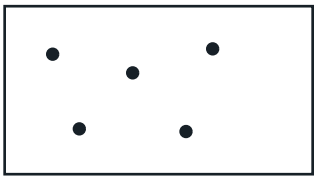
Word help: ideal = using stated simplifying conditions

Sentence starter: Both ... . They differ in ... because ... .

Sample A



Sample B



Add velocity arrows; use lengths to show a range of speeds.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

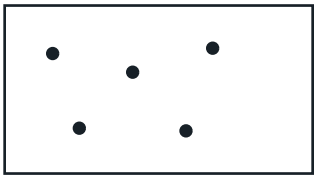
04 Apply the physics

An ideal-gas simulation warms a sample from **300 K** to **600 K**. By what factor does the average translational kinetic energy per particle change? **Label the two particle diagrams** to **show your comparison**. Distinguish energy from speed.

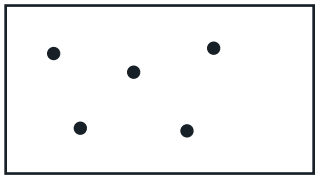
Word help: ideal = using stated simplifying conditions

Sentence starter: In my work, I used … to show … . I labeled it … .

Sample A



Sample B



Add velocity arrows; use lengths to show a range of speeds.

Marzano check - circle: ① ② ③

05 Explain your reasoning

A student doubles **20°C** to **40°C** and says average particle kinetic energy doubles. **Explain why the temperature scale** matters.

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

Two students perform the same lift. One takes **4** times as long as the other. **Compare work** and average power. **Use the table** to distinguish energy from its transfer rate.

Word help: power = rate of energy transfer

Sentence starter: Both … . They differ in … because … .

Case	Energy (J)	Time (s)	Power (W)
A	____	____	____
B	____	____	____

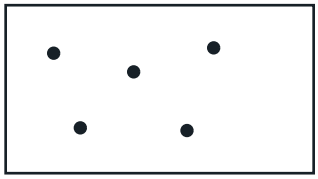
Marzano check - circle: ① ② ③

07 Exit check

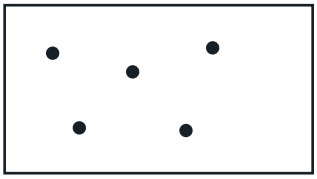
At equal kelvin temperature, does a gas sample with twice as many particles have twice the average energy per particle?

Sentence starter: I notice … . This supports … because … .

Sample A



Sample B



Add velocity arrows; use lengths to show a range of speeds.

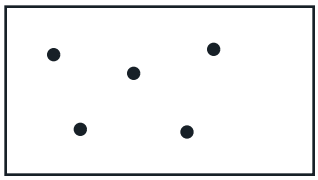
Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

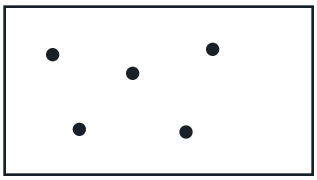
Two gas samples have the same temperature but different volumes. Must their particle speeds differ? **Explain what temperature alone tells** you in this ideal-gas example.

Sentence starter: Both … . They differ in … because … .

Sample A



Sample B



Add velocity arrows; use lengths to show a range of speeds.

Marzano check - circle: ① ② ③